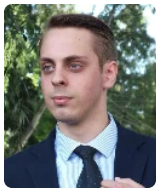
Petar Lukovic

COMPUTER VISION · MSC ELECTRICAL ENGINEERING, ETH ZÜRICH

Zürich, Switzerland petar.lukovic032@gmail.com LinkedIn Website



PROFESSIONAL SUMMARY

Enthusiastic and highly driven Electrical Engineering graduate (MSc, ETH Zürich), motivated to deepen expertise in Computer Vision, 3D Vision and Robotics. Academic experience in Computer Vision and 3D Vision, combined with an outgoing and communicative personality that thrives in group projects and teams. Seeking opportunities in these fields.

EDUCATION

Sept 2023 — Present
Master's, ETH Zürich — Zürich
- Department of Electrical Engineering, specialization in Signal Processing and Machine Learning

Oct 2019 — Jul 2023
Bachelor's, University of Belgrade — Belgrade
- School of Electrical Engineering, Department for Signals and Systems

PROJECTS AND PUBLICATIONS

Mar 2026 — Sept 2026
Polygraph: Large-Scale Multi-Session Trajectory Optimization Using Pose Graphs. Master Thesis — ETH Zürich
- Mapless multi-session trajectory optimization with pose graphs and learned metric depth. Under submission @ ICRA 2027

Mar 2025 — Present
CroCoDL: Cross-device Collaborative Dataset for Localization — ETH Zürich
- Data processing and maintenance. Challenge organiser @ ICCV 2025. Website Challenge

Aug 2025 — Dec 2025
Interactive 3D Segmentation. Semester project — ETH Zürich
- Interactive 3D Segmentation using Segment Anything

Oct 2024 — Mar 2025
Spot-On: A Mixed Reality Interface for Multi-Robot Cooperation — ETH Zürich
- Intuitive mixed reality interface for human and robot collaboration. CV4MR Workshop @ CVPR 2025. Paper

Sept 2024 — Mar 2025
HoloSpot: Intuitive Object Manipulation via Mixed Reality Drag-and-Drop — ETH Zürich
- Integration of Boston Dynamics Spot robot and Microsoft HoloLens 2 for Drag and Drop tasks with help of OpenMask3D and AnyGrasp. Website

Mar 2024 — Aug 2024
Multiple to One Matching. Semester project — ETH Zürich
- Matching multiple database images to one query image for camera localization and 3D reconstruction using HLoc

Mar 2023 — Jul 2023
Stereo Vision Depth Estimation. Bachelor Thesis — University of Belgrade
- Theoretical basis for Stereo Depth estimation. Practical implementation on Nvidia Jetson Nano. Thesis

EXTRACURRICULAR ACTIVITIES

Mar 2025 — Present
Research Assistant, ETH Zürich — Zürich
- Research Assistant at Computer Vision and Geometry Lab

Sept 2025 — Feb 2026
Teaching Assistant, ETH Zürich — Zürich
- Teaching Assistant on Mixed Reality (HS25) at Computer Vision and Geometry Lab

Sept 2024 — Feb 2026
Teaching Assistant, ETH Zürich — Zürich
- Teaching Assistant on Computer Vision (HS24) at Computer Vision and Geometry Lab

Oct 2022 — Jun 2023
Teaching Assistant, School of Electrical Engineering — Belgrade
- Teaching Assistant on Object-oriented Programming (HS20) and Microcontroller Systems (HS22)

Oct 2021 — May 2023
Team member, Formula Student team Road Arrow — Belgrade
- Member of the team for vehicle dynamics and powertrain

LANGUAGES

English C2 Serbian Native

SKILLS

Python, C / C++, CUDA
Unity, Linux, Git
SLAM, Robotics, Computer Vision, 3D Vision
HoloLens 2, Meta Quest 3, Aria Gen 2
Boston Dynamics Spot